ELECTRODYNAMICS II · SEMESTER 5

Transients and alternating currents

Module 4 · Prepare

Start with the idea. Learn the mathematics. Work the questions. Check the result.

CU-FYUGP syllabus 2024 · study edition
SI units · phone pages · October 2026

How to use this book

P means prerequisite, M means mathematics, T means topic and Q means the numbered practice question. Topic codes restart in each module. Read P1 and M1 first, then follow the topics in order.

REMIND ME

A fact from school that we will use again.

MATHS TOOL

The meaning of a symbol and the operation it asks you to do.

THE IDEA

A physical explanation before the equation.

RESULT

The statement to carry into an exam.

EXAM TRAP

An assumption, sign or unit that changes the answer.

CHECK YOURSELF

Cover the answer and try. Then explain why your answer makes sense.

One rule for this module

> **KEEP THIS IN MIND**
>
> Write the circuit law with a consistent sign convention, then decide whether you are studying a transient or a sinusoidal steady state.

The whole module in five sentences

1. A resistor dissipates energy, a capacitor stores electric energy and an inductor stores magnetic energy.
2. After a DC switch changes position, stored quantities approach their new values through a transient.
3. An ideal LC circuit exchanges energy between its two stores and oscillates rather than settling.
4. Sinusoidal steady-state circuits are simpler when voltage and current are represented by phasors.
5. At resonance reactances cancel, while average power is determined by the in-phase part of current.

Syllabus route

Units 15–22 · 13 hours · 18 theory marks. Murugeshan, Electricity and Magnetism, 10th ed., §§12.1, 12.3, 12.5, 13.1, 13.2, 13.3, 13.5. LR decay and parallel LC are explicitly in the syllabus; related reading is §§12.2 and 13.4.

Definitions and derivations are independently explained from the named textbook sections. Worked numerical values are original practice unless a past-paper tag is printed.

Topic route

Units 15–16 → T1–T5; unit 17 → T6; unit 18 → T7; unit 19 → T8; unit 20 → T8–T9; unit 21 → T10; unit 22 → T11.

T1 · LR current growth

T2 · LR current decay

T3 · RC charging and discharging

T4 · Free LC oscillations

T5 · LC response to a DC step

T6 · Rotating-coil AC generator

T7 · Pure R, L and C on AC

T8 · Complex impedance and series RL or RC

T9 · Parallel LC and antiresonance

T10 · Series LCR and resonance

T11 · Power in AC

P1 School physics bridge

Three elements, three jobs

A resistor obeys $v_R = Ri$ and turns electrical energy into heat. A capacitor obeys $q = Cv_C$ and stores $U_C = q^2/(2C)$. An ideal inductor obeys $v_L = L\,di/dt$ and stores $U_L = Li^2/2$. R is in ohms, C in farads, L in henries.

Kirchhoff before algebra

At a node, incoming current equals outgoing current: charge does not pile up in the ideal wire junction. Around a loop the sum of voltage rises and drops is zero. Draw a current arrow first. A passive voltage drop is positive from the terminal where that chosen current enters to the terminal where it leaves.

What can change instantly?

With finite voltage, inductor current is continuous because an instantaneous change would need unbounded $L\,di/dt$. With finite current, capacitor voltage is continuous because $i = C\,dv_C/dt$. A resistor has no stored state and can change its voltage/current immediately in the ideal model.

DC and AC are different questions

A transient asks what happens just after switching and includes initial conditions. Sinusoidal steady state asks what remains after any decaying transient has died away. For ideal DC steady state

an inductor acts as a short circuit and a capacitor as an open circuit; these are not general statements at switching time.

Prefixes

$1\ \mathrm{mH} = 10^{-3}$ H, $1\ \mu\mathrm{F} = 10^{-6}$ F, and $1\ \mathrm{ms} = 10^{-3}$ s. Multiply L and C in SI units before taking a square root.

M1 Mathematics from the beginning

Exponentials and the time constant

The number $e \simeq 2.718$ has a special exponential function: $d(e^x)/dx = e^x$. Thus $d(e^{-t/\tau})/dt = -e^{-t/\tau}/\tau$. The exponent is dimensionless, so τ must be a time. At t=τ, the exponential is about 0.368; a growth curve $1 - e^{-t/\tau}$ has reached 0.632 of its final value.

Solving a first-order equation

For $dy/dt = (y_\infty - y)/\tau$, put $w = y_\infty - y$. Then $dw/dt = -w/\tau$. Separate: $dw/w = -dt/\tau$. Integration gives $\ln w = -t/\tau + \text{constant}$, or $w = w_0 e^{-t/\tau}$. Hence $y = y_\infty + (y_0 - y_\infty)e^{-t/\tau}$. A logarithm reverses an exponential.

Oscillations are second-order

$d^2(\cos \omega t)/dt^2 = -\omega^2 \cos \omega t$. Therefore $d^2q/dt^2 + \omega_0^2 q = 0$ is solved by a cosine plus a sine. Two initial conditions, q(0) and i(0), fix the two

constants. A spring's displacement obeys the same form; an LC circuit swaps stored energy instead.

The j operator

Define $j^2 = -1$. A complex number $a + jb$ is drawn as a point with horizontal a and vertical b. Multiplying by j rotates it +90°. Its magnitude is $\sqrt{a^2 + b^2}$ and its angle satisfies $\tan\phi = b/a$, with the quadrant checked.

Euler's formula $e^{j\theta} = \cos\theta + j\sin\theta$ packages a sinusoid. Represent $v(t) = \mathrm{Re}[\widetilde{V}e^{j\omega t}]$; the complex coefficient is a phasor. Differentiation becomes multiplication by jω. We use **rms phasor magnitudes** in circuit calculations; time-domain peak is $\sqrt{2}$ times rms.

Adding and dividing phasors

Series impedances add. Parallel admittances $Y = 1/Z$ add. To divide by $a + jb$, multiply top and bottom by $a - jb$: $(a + jb)(a - jb) = a^2 + b^2$. Example: $1/(3 + j4) = (3 - j4)/25$, with magnitude 1/5.

Cycle averages

$\langle \sin^2 \omega t \rangle = 1/2$. Also $\langle \cos\omega t \cos(\omega t - \phi) \rangle = \cos\phi/2$. The oscillating double-frequency term averages to zero. This is why rms values and a power factor appear in average power.

T1 LR current growth

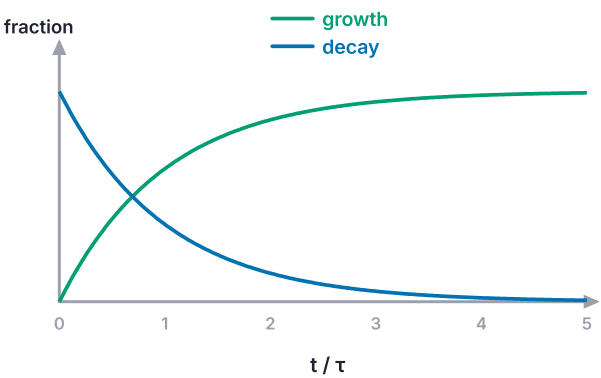
> Purpose: An inductor initially opposes the change of current; resistance sets the final DC value.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

An inductor initially opposes the change of current; resistance sets the final DC value.



Computed exponential growth and decay. At t/τ=1 the fractions are 0.632 and 0.368; the same time scale describes the remaining gap.

1. Kirchhoff gives $L\,di/dt + Ri = V$.
   Why: The source balances inductive and resistive drops.

2. The final derivative is zero, giving $i_\infty = V/R$.
   Why: At steady DC there is no changing flux.

3. The remaining distance to that final value decays exponentially with time L/R.
   Why: The first-order equation has a constant coefficient.

4. At t=0, i=0 and $v_L = V$; at long time, $v_L = 0$.
   Why: The current cannot jump but its derivative can.

RESULT

$i = (V/R)(1 - e^{-t/\tau_L})$, $\tau_L = L/R$, for i(0)=0 and a DC step V.

EXAM TRAP

Increasing R decreases the time constant and also decreases the final current at fixed V.

CHECK YOURSELF

At one LR time constant, what fraction of final current flows?
Answer: $1 - e^{-1} \simeq 0.632$.

This topic supports Q1, Q2, Q21, Q22.

Work it through · Q1

Skill: LR time constant. Model · syllabus-based

Define the LR time constant, state its unit and explain its growth interpretation.

$\tau_L = L/R$ has units H/Ω = s. After one τ, initially zero current has risen to $1 - e^{-1} = 0.632$ of V/R. Equivalently the gap remaining to its final value is e^{-1} of its original gap.

CHECK THE ENDPOINT

$\tau_L = L/R$; 63.2% rise in one time constant.

Work it through · Q2

Skill: Current continuity at switching. Model · syllabus-based

Why does current in an ideal inductor remain continuous when a finite DC voltage is suddenly applied?

The passive voltage is $v_L = L\,di/dt$. A finite voltage gives a finite slope, not an instantaneous finite jump. The initial current is therefore the pre-switch value even though the applied voltage changes abruptly.

CHECK THE ENDPOINT

Finite inductor voltage implies continuous current.

Work it through · Q21

Skill: LR step solution. Model · syllabus-based

Derive current growth in a series LR circuit connected to constant V at t=0, with i(0)=0.

1. Kirchhoff gives $L\dot{i} + Ri = V$.
   Why: Source voltage equals the two passive drops.

2. Write $\dot{i} = (R/L)(V/R - i)$.
   Why: This exposes the final value and decay coefficient.

3. The general solution is $i = V/R + Ke^{-Rt/L}$.
   Why: The deviation from the final value satisfies a decay equation.

4. At t=0, $0 = V/R + K$, so K=−V/R.
   Why: Apply the initial inductor current.

5. Check i(0)=0, i(∞)=V/R and initial slope V/L.
   Why: All three physical limits agree.

CHECK THE ENDPOINT

$i(t) = (V/R)(1 - e^{-Rt/L})$.

Work it through · Q22

Skill: LR transient numerical. Model · syllabus-based

A 12 V step drives R=6 Ω and L=3 H from zero current. Find τ, i and inductor voltage at t=0.50 s.

1. $\tau = L/R = 0.50$ s and $I_\infty = V/R = 2$ A.
   Why: Determine time and current scales first.

2. At one τ, $i = 2(1 - e^{-1}) = 1.264$ A.
   Why: Use the growth solution.

3. $v_L = V - Ri = 12 - 6(1.264) = 4.415$ V.
   Why: The source voltage is shared between R and L.

4. Equivalently $v_L = Ve^{-1}$.
   Why: The inductor's remaining drop decays exponentially.

CHECK THE ENDPOINT

τ=0.50 s; i≈1.264 A; $v_L \approx 4.415$ V.

T2 LR current decay

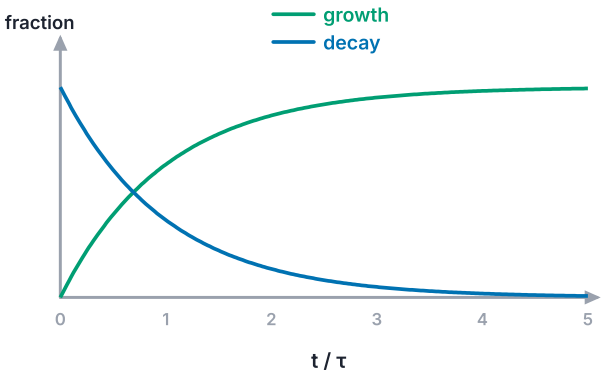
Purpose: After removing the source, an existing inductor current continues through the closed resistor path and dies away.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

After removing the source, an existing inductor current continues through the closed resistor path and dies away.



Computed exponential growth and decay. At t/τ=1 the fractions are 0.632 and 0.368; the same time scale describes the remaining gap.

1. Choose the discharge loop so current initially retains its original direction.
   Why: Inductor current is continuous.
2. Kirchhoff gives $L\,di/dt + Ri = 0$.
   Why: There is no source supplying a rise.
3. Solve with i(0)=I_0 to obtain exponential decay.
   Why: The derivative is proportional to negative current.
4. Stored field energy becomes resistor heat.
   Why: An ideal inductor has no internal loss here.

RESULT

$i = I_0 e^{-tR/L}$; $U_L = LI_0^2 e^{-2tR/L}/2$.

EXAM TRAP

Simply opening the current path can create a large voltage; the ideal decay formula assumes a closed R-L path.

CHECK YOURSELF

When current has fallen to half, what fraction of magnetic energy remains?
Answer: One quarter.

This topic supports Q3, Q4, Q23, Q24.

Work it through · Q3

Skill: LR half-current energy. Model · syllabus-based

During LR decay, how much magnetic energy remains when the current is half its initial value?

$U_L = Li^2/2$. Therefore $U/U_0 = (i/I_0)^2 = (1/2)^2 = 1/4$. The remaining three quarters has been dissipated in the resistor for the ideal closed discharge path.

CHECK THE ENDPOINT

25% remains.

Work it through · Q4

Skill: Discharge path condition. Model · syllabus-based

Why must the LR decay formula specify a closed discharge path?

The formula solves $L\dot{i} + Ri = 0$ with a finite resistor path. If a switch simply opens an inductor carrying current, a large voltage can develop to

maintain continuity, potentially producing arcing. That situation is not the assumed ideal R-L loop.

CHECK THE ENDPOINT

A defined resistor path is part of the exponential-decay model.

Work it through · Q23

Skill: LR decay solution and heat. Model · syllabus-based

Derive LR decay from initial current I_0 and show the total resistor heat equals the initial magnetic energy.

1. $L\dot{i} + Ri = 0$ gives $di/i = -(R/L)dt$.
   Why: No source remains in the closed loop.
2. Integrate and impose i(0)=I_0: $i = I_0 e^{-Rt/L}$.
   Why: The constant fixes the initial current.
3. Heat is $W = \int_0^\infty Ri^2 dt = RI_0^2 \int_0^\infty e^{-2Rt/L} dt$.
   Why: Joule power is Ri^2.
4. The integral equals L/(2R), giving $W = LI_0^2/2$.
   Why: The exponential vanishes at infinity.

CHECK THE ENDPOINT

All initial magnetic energy becomes resistor heat: $W = LI_0^2/2$.

Work it through · Q24

Skill: Decay timing with logarithm. Model · syllabus-based

An LR circuit has L=2 H and R=10 Ω. How long does current take to fall to 10% of its initial value?

1. $i/I_0 = e^{-t/\tau}$ with $\tau = L/R = 0.20$ s.
   Why: Use the current decay time, not the energy decay time.
2. Set $0.10 = e^{-t/0.20}$ and take logarithms: $\ln 0.10 = -t/0.20$.
   Why: A logarithm reverses the exponential.
3. $t = 0.20 \ln 10 = 0.4605$ s.
   Why: The negative log becomes a positive elapsed time.

CHECK THE ENDPOINT

Current reaches 10% at about 0.461 s.

T3 RC charging and discharging

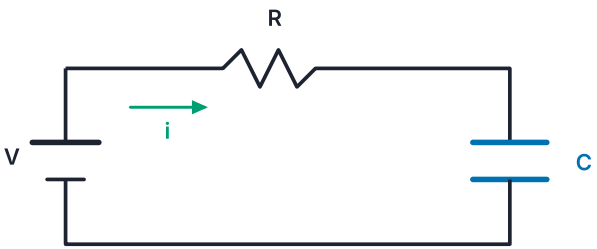
> Purpose: A resistor limits how quickly a capacitor's voltage changes.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

A resistor limits how quickly a capacitor's voltage changes.



Charging loop: V = Ri + q/C

Series RC charging with the switch understood to close at t=0. The initially positive capacitor plate is the upper plate.

1. Use $Ri + q/C = V$ and $i = dq/dt$.
   Why: Kirchhoff plus charge storage gives a first-order equation.

2. Initially an uncharged capacitor has zero voltage; after charging it holds the source voltage.
   Why: Capacitor voltage is continuous at switching.

3. Remove the source into a closed R-C path and set the right side to zero.
   Why: Stored electric energy drives discharge.

4. The negative discharge current means q decreases in the chosen capacitor-charge convention.
   Why: Current signs depend on the arrow definition.

RESULT

Charging: $q = CV(1 - e^{-t/RC})$, $i = (V/R)e^{-t/RC}$. Discharging: $q = q_0 e^{-t/RC}$, $i = -q_0 e^{-t/RC}/(RC)$ with i=dq/dt.

EXAM TRAP

Charge grows during charging while current decays; do not copy the same time dependence for both.

CHECK YOURSELF

At t=RC during charging, are q and i at the same fraction of their initial/final scales?
Answer: No: q is 0.632 of final; i is 0.368 of initial.

This topic supports Q5, Q6, Q25, Q26, Q27, Q37.

Work it through · Q5

Skill: RC time constant. Model · syllabus-based

An RC circuit has R=20 kΩ and C=50 μF. Find τ and the charging fraction at t=τ.

$\tau = RC = (2.0 \times 10^4)(50 \times 10^{-6}) = 1.0$ s. The capacitor voltage and charge reach $1 - e^{-1} = 0.632$ of their final values, while charging current falls to 0.368 of its initial value.

CHECK THE ENDPOINT

τ=1.0 s; charge fraction 63.2%.

Work it through · Q6

Skill: Capacitor continuity. Model · syllabus-based

Which capacitor quantity remains continuous under finite current at switching, and why?

Capacitor voltage remains continuous because $i = C\,dv_C/dt$. A finite instantaneous voltage jump would need an impulse of unbounded current in the ideal model. Charge q=Cv is likewise continuous.

CHECK THE ENDPOINT

Finite current implies continuous capacitor voltage and charge.

Work it through · Q25

Skill: RC charge solution. Model · syllabus-based

Derive q(t), i(t) and capacitor voltage when an initially uncharged series RC circuit is connected to DC V.

1. $R\dot{q} + q/C = V$.
   Why: Use Kirchhoff and i=dq/dt.
2. The final charge is CV and the time constant is RC.
   Why: Compare to the standard first-order equation.
3. $q = CV(1 - e^{-t/RC})$ meets q(0)=0.
   Why: The deviation from CV decays exponentially.

4. Differentiate to get $i = (V/R)e^{-t/RC}$ and divide q by C for $v_C = V(1 - e^{-t/RC})$.
   Why: Current is charge rate and voltage is charge per capacitance.

5. Initially i=V/R while $v_C = 0$; finally i=0 while $v_C = V$.
   Why: These limits check the solution.

CHECK THE ENDPOINT

Charge and voltage grow; charging current decays.

Work it through · Q26

Skill: RC discharge sign and numerical. Model · syllabus-based

A 100 µF capacitor initially at 10 V discharges through 20 kΩ. Using i=dq/dt for its initially positive plate, find q, v and i after 2 s.

1. $\tau = RC = 2$ s and $q_0 = CV_0 = 1.0 \times 10^{-3}$ C.
   Why: Convert microfarads to farads.

2. $q = q_0 e^{-1} = 3.679 \times 10^{-4}$ C and $v = 10e^{-1} = 3.679$ V.
   Why: The source-free charge decays.

3. $i = -q_0 e^{-1}/\tau = -1.839 \times 10^{-4}$ A.
   Why: The chosen positive plate loses charge.

4. A current arrow in the discharge direction would instead have positive magnitude 0.1839 mA.
   Why: A negative sign describes the reference convention.

CHECK THE ENDPOINT

q≈0.3679 mC; v≈3.679 V; i≈−0.1839 mA with i=dq/dt.

Work it through · Q27

Skill: RC charging energy budget. Model · syllabus-based

For charging an initially uncharged capacitor from an ideal constant source V through R, find source work, final stored energy and resistor heat.

1. Total charge supplied is CV, so source work is $W_s = V\int i\,dt = V(CV) = CV^2$.
   Why: The source voltage stays constant throughout charging.

2. Final stored energy is $U_C = CV^2/2$.
   Why: Capacitor energy is the integral of its rising voltage over charge.

3. The difference $W_R = W_s - U_C = CV^2/2$ is resistor heat.
   Why: Energy is conserved.

4. The result is independent of positive R, although R changes the charging duration.
   Why: A smaller R raises current but shortens the transient.

CHECK THE ENDPOINT

Half the source energy is stored; half is dissipated in this ideal resistive charging process.

Work it through · Q37

Skill: Comparing first-order transients. Model · syllabus-based

Derive and compare LR growth/decay and RC charge/discharge, emphasizing initial conditions, time constants and limiting behaviour.

1. LR obeys $L\dot{i} + Ri = V$; RC obeys $R\dot{q} + q/C = V$.
   Why: Use Kirchhoff with the element relations.

2. Each has form $\dot{y} = (y_\infty - y)/\tau$ and solution $y = y_\infty + (y_0 - y_\infty)e^{-t/\tau}$.
   Why: A deviation from steady state decays exponentially.

3. For LR, $\tau = L/R$ and $i_\infty = V/R$; for RC, $\tau = RC$ and $q_\infty = CV$.
   Why: Compare coefficients.

4. Removing the source into a closed discharge path sets the final state to zero, giving $i = I_0 e^{-tR/L}$ or $q = q_0 e^{-t/RC}$.
   Why: Stored energy drives a decaying response.

5. Inductor current and capacitor voltage remain continuous at switching under finite drive.
   Why: Their derivatives are limited by finite voltage or current.

6. During RC charging, current is $Ve^{-t/RC}/R$, while charge grows; during LR growth, current itself rises.
   Why: Current equals charge derivative only in the capacitor state equation.

7. Check t=0, t=∞ and units of τ in every solution.
   Why: These three checks catch sign and reciprocal errors.

CHECK THE ENDPOINT

LR time scale is L/R; RC time scale is RC; both are first-order responses with specified initial states.

T4 Free LC oscillations

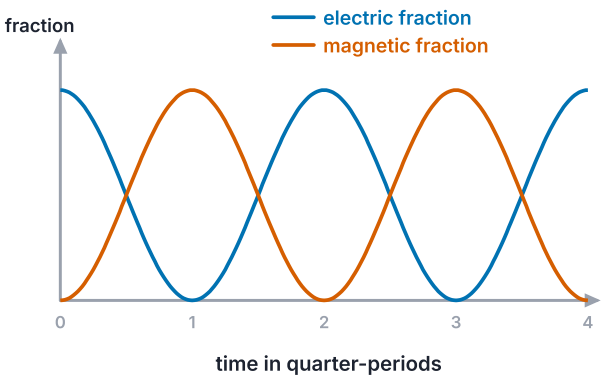
> Purpose: An ideal capacitor and inductor swap electric and magnetic energy indefinitely.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

An ideal capacitor and inductor swap electric and magnetic energy indefinitely.



Computed free-LC energy fractions: electric plus magnetic energy is one at every instant.

1. Kirchhoff in the isolated loop gives $L\,di/dt + q/C = 0$.
   Why: There is no resistor or source.
2. Use i=dq/dt to get a second-order equation.
   Why: Capacitor charge is the state variable.
3. At maximum charge current is zero; a quarter-cycle later charge is zero and current magnitude is greatest.
   Why: The energy moves between C and L.
4. The sum $q^2/(2C) + Li^2/2$ stays constant.
   Why: There is no dissipative element.

RESULT

$\ddot{q} + q/(LC) = 0$; $\omega_0 = 1/\sqrt{LC}$; for q(0)=q_0, i(0)=0: $q = q_0 \cos \omega_0 t$, $i = -\omega_0 q_0 \sin \omega_0 t$.

EXAM TRAP

An ideal LC circuit has no exponential decay. Real resistance makes the oscillation damped.

CHECK YOURSELF

When q=0 in a free LC oscillation, has all circuit energy vanished?

Answer: No. It is stored in the inductor.

This topic supports Q7, Q8, Q28, Q38.

Work it through · Q7

Skill: LC natural frequency. Model · syllabus-based

Write the natural angular frequency and frequency of an ideal LC circuit, and distinguish the units.

$\omega_0 = 1/\sqrt{LC}$ in rad/s, while $f_0 = \omega_0/(2\pi) = 1/(2\pi\sqrt{LC})$ in hertz. Confusing the two loses a factor of 2π.

CHECK THE ENDPOINT

Angular frequency and ordinary frequency differ by 2π.

Work it through · Q8

Skill: Quarter-cycle energy exchange. Model · syllabus-based

An isolated LC circuit starts with a charged capacitor and zero current. Where is its energy after a quarter-cycle?

At $t = T/4$, $q = q_0 \cos(\pi/2) = 0$, so capacitor energy is zero. Current magnitude is maximal and all energy is magnetic, $Li^2/2 = q_0^2/(2C)$.

CHECK THE ENDPOINT

All stored energy is in the inductor.

Work it through · Q28

Skill: Free LC solution and numerical. Model · syllabus-based

An ideal isolated circuit has L=0.10 H, C=10 μF, initial capacitor voltage 20 V and zero current. Find q(t), i(t), frequency and peak current.

1. $q_0 = CV_0 = 2.0 \times 10^{-4}$ C and $\omega_0 = 1/\sqrt{LC} = 1000$ rad/s.
   Why: Use SI units in LC.

2. $q = q_0 \cos 1000t$ and $i = -\omega_0 q_0 \sin 1000t$.
   Why: Apply initial q and zero initial current.

3. Thus $i = -0.20 \sin 1000t$ A.
   Why: Current is dq/dt.

4. $f_0 = 1000/(2\pi) = 159.15$ Hz.
   Why: Convert angular to ordinary frequency.

CHECK THE ENDPOINT

$q = 2.0 \times 10^{-4} \cos 1000t$ C, $i = -0.20 \sin 1000t$ A; peak current 0.20 A.

Work it through · Q38

Skill: Free versus driven LC and conservation. Model · syllabus-based

Develop free LC oscillation and compare it with an initially uncharged LC circuit subjected to a DC step. Explain energy in both cases.

1. Free LC gives $L\ddot{q} + q/C = 0$, so for q(0)=q_0 and i(0)=0, $q = q_0 \cos \omega_0 t$ and $i = -\omega_0 q_0 \sin \omega_0 t$.
   Why: The natural frequency is $1/\sqrt{LC}$.

2. Substituting into $U = q^2/(2C) + Li^2/2$ gives $U = q_0^2/(2C)$.
   Why: Sine and cosine squared sum to one and $L\omega_0^2 = 1/C$.

3. Energy alternates between C and L without loss.
   Why: No resistance or external source is present.

4. For a DC step the equation is $L\ddot{q} + q/C = V$, with zero initial q and i.
   Why: The source remains connected and changes the forcing.

5. The solution is $q = CV(1 - \cos \omega_0 t)$ and $i = CV\omega_0 \sin \omega_0 t$.
   Why: Shift q by CV and apply initial conditions.

6. Now the stored energy changes by source power Vi: $dU/dt = Vi$.
   Why: Multiply the circuit equation by i.

7. At half a cycle q=2CV, stored energy is $2CV^2$, exactly source work $Vq = 2CV^2$; on the return the ideal source can receive energy.
   Why: Overshoot is consistent with conservation.

CHECK THE ENDPOINT

Free LC has constant total stored energy; the driven LC step exchanges energy with the source and does not settle.

T5 LC response to a DC step

Purpose: A constant source connected to an ideal LC series circuit does not create a settling exponential.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

A constant source connected to an ideal LC series circuit does not create a settling exponential.

1. Write $L\ddot{q} + q/C = V$.
   Why: The source remains in the loop.

2. Subtract the equilibrium charge CV: set x=q−CV.
   Why: This turns the equation into $\ddot{x} + \omega_0^2 x = 0$.

3. Initial conditions are x(0)=−CV and $\dot{x}(0) = 0$.
   Why: The capacitor was uncharged and the inductor carried no current.

4. Solve to obtain q and i; capacitor voltage oscillates between 0 and 2V.
   Why: Without resistance there is no mechanism to settle.

RESULT

For initially uncharged C and zero current: $q = CV(1 - \cos \omega_0 t)$, $i = CV\omega_0 \sin \omega_0 t$.

EXAM TRAP

This driven step response differs from free LC discharge; the source exchanges energy with the circuit.

CHECK YOURSELF

Does an ideal LC step current approach a nonzero constant?
Answer: No. It oscillates about zero.

This topic supports Q9, Q29.

Work it through · Q9

Skill: Ideal LC step overshoot. Model · syllabus-based

An uncharged capacitor and ideal inductor are connected in series to a DC source V at t=0. What is the maximum capacitor voltage?

$v_C = V(1 - \cos \omega_0 t)$ under zero initial current and charge. At half a natural period, cosine is −1 and $v_C = 2V$. The ideal circuit does not settle because it lacks resistance.

CHECK THE ENDPOINT

Maximum capacitor voltage is 2V.

Work it through · Q29

Skill: Driven ideal LC step derivation. Model · syllabus-based

Derive the ideal series LC response to a DC step V with zero initial charge and current. Explain the voltage overshoot.

1. $L\ddot{q} + q/C = V$.
   Why: The source remains connected.
2. Set x=q−CV to obtain $\ddot{x} + \omega_0^2 x = 0$.
   Why: The constant forcing shifts the centre of oscillation.
3. Since x(0)=−CV and $\dot{x}(0) = 0$, $x = -CV \cos \omega_0 t$.
   Why: Use both initial conditions.
4. Therefore $q = CV(1 - \cos \omega_0 t)$ and $i = CV\omega_0 \sin \omega_0 t$.
   Why: Differentiate charge for current.

5. At $\omega_0 t = \pi$, $v_C = 2V$ and i=0.
   Why: The ideal inductor's stored energy and source work drive the overshoot.

6. No resistor removes the oscillation energy.
   Why: The circuit does not settle in this ideal model.

CHECK THE ENDPOINT

The capacitor voltage oscillates from 0 to 2V; this is distinct from free discharge.

T6 Rotating-coil AC generator

Purpose: Rotation changes flux linkage sinusoidally and produces an alternating emf.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

Rotation changes flux linkage sinusoidally and produces an alternating emf.

1. The linked flux is $NBA\cos\omega t$.
   Why: Area and field are fixed while orientation changes.

2. Differentiate and apply Faraday's minus sign.
   Why: The emf is the negative flux-linkage rate.

3. One complete rotation gives one sinusoidal cycle for this ideal two-pole geometry.
   Why: The orientation repeats after 2π radians.

RESULT

$\mathcal{E} = NBA\omega \sin \omega t$, peak $\mathcal{E}_0 = NBA\omega$, when the coil normal starts along B.

EXAM TRAP

Maximum flux occurs at zero emf; maximum emf occurs where flux crosses zero.

CHECK YOURSELF

If angular speed doubles, how do frequency and peak emf change?
Answer: Both double.

This topic supports Q10, Q11, Q30.

Work it through · Q10

Skill: Generator peak dependence. Model · syllabus-based

State the peak emf of a rotating N-turn coil and explain its dependence on angular speed.

For area A in uniform B, $\mathcal{E}_0 = NBA\omega$. Faster rotation increases the rate of change of linked flux, so peak emf is proportional to ω. The result assumes rotation about an axis in the coil plane perpendicular to B.

CHECK THE ENDPOINT

$\mathcal{E}_0 = NBA\omega.$

Work it through · Q11

Skill: Flux-emf timing. Model · syllabus-based

At what coil orientations are flux and emf maximum in an ideal AC generator?

Flux magnitude is maximum when the coil normal is parallel or antiparallel to B; its derivative and emf are then zero. Emf magnitude is maximum when the normal is perpendicular to B, where flux crosses zero fastest.

CHECK THE ENDPOINT

Flux and emf maxima are separated by a quarter-cycle.

Work it through · Q30

Skill: Generator numerical. Model · syllabus-based

A 100-turn coil of area 0.020 m² rotates at 50 revolutions/s in a 0.10 T uniform field. Find angular speed, emf peak and rms emf.

1. $\omega = 2\pi f = 100\pi$ rad/s.
   Why: One revolution is 2π radians.

2. $\mathcal{E}_0 = NBA\omega = 100(0.10)(0.020)(100\pi) = 62.83$ V.
   Why: Differentiate the rotating flux linkage.

3. $\mathcal{E}_{\text{rms}} = \mathcal{E}_0/\sqrt{2} = 44.43$ V.
   Why: The emf is sinusoidal.

4 The electrical frequency here is 50 Hz.
Why: This is the ideal two-pole geometry stated in the topic.

CHECK THE ENDPOINT

ω=314.16 rad/s; peak emf 62.83 V; rms emf 44.43 V.

T7 Pure R, L and C on AC

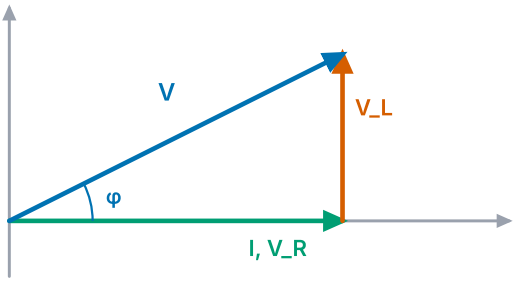
Purpose: Resistance keeps voltage and current in phase; reactive elements shift them by a quarter-cycle.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

Resistance keeps voltage and current in phase; reactive elements shift them by a quarter-cycle.



Series RL phasors with current as reference. V is the vector sum of V_R and V_L, so the source voltage leads current.

1. For R, v=Ri, so their waveforms align.
   Why: No derivative or integral shifts phase.
2. For L, $v = L\,di/dt$: voltage leads current by 90°.
   Why: A derivative advances sinusoidal phase.
3. For C, $i = C\,dv/dt$: current leads voltage by 90°.
   Why: Here current is the derivative.
4. Ideal L and C exchange energy with the source but have zero mean power.
   Why: Their instantaneous power alternates sign equally.

RESULT

$Z_R = R$, $Z_L = j\omega L$, $Z_C = 1/(j\omega C) = -j/(\omega C)$.

EXAM TRAP

“ELI the ICE man”: in L, E leads I; in C, I leads E. State which quantity leads.

CHECK YOURSELF

What happens to capacitive reactance when frequency doubles?

Answer: It halves.

This topic supports Q12, Q13, Q14, Q31.

Work it through · Q12

Skill: Pure resistor phase. Model · syllabus-based

Describe voltage, current and average power for a pure resistor driven by sinusoidal voltage.

Current is in phase with voltage because $i = v/R$. $I_{\text{rms}} = V_{\text{rms}}/R$ and $P = V_{\text{rms}} I_{\text{rms}} = I_{\text{rms}}^2 R$. Instantaneous power is nonnegative.

CHECK THE ENDPOINT

A resistor dissipates positive mean power with power factor one.

Work it through · Q13

Skill: Pure inductor phase. Model · syllabus-based

State the reactance and phase relation for an ideal inductor on sinusoidal AC.

$X_L = \omega L$ and $Z_L = j\omega L$. Voltage leads current by 90°, or current lags voltage by 90°. Average power is zero because stored magnetic energy is returned each cycle.

CHECK THE ENDPOINT

Inductive current lags voltage by 90°.

Work it through · Q14

Skill: Pure capacitor phase. Model · syllabus-based

State the reactance and phase relation for an ideal capacitor on sinusoidal AC.

$X_C = 1/(\omega C)$ as a positive magnitude and $Z_C = -jX_C$. Current leads voltage by 90° because $i = C\,dv/dt$. Mean power is zero for an ideal capacitor.

CHECK THE ENDPOINT

Capacitive current leads voltage by 90°.

Work it through · Q31

Skill: Deriving reactive impedances. Model · syllabus-based

Derive the impedances of an ideal inductor and capacitor using j and the common time factor $e^{j\omega t}$.

1. Differentiation of a sinusoidal phasor expression multiplies it by jω.
   Why: The derivative of $e^{j\omega t}$ is $j\omega e^{j\omega t}$.
2. For L, $\widetilde{V} = j\omega L\tilde{I}$, giving $Z_L = j\omega L$.
   Why: Use $v = L\,di/dt$.
3. For C, $\tilde{I} = j\omega C\widetilde{V}$, giving $Z_C = 1/(j\omega C) = -j/(\omega C)$.
   Why: Use $i = C\,dv/dt$ and $1/j = -j$.
4. Their angles are +90° and −90° respectively.
   Why: Multiplication by ±j rotates the phasor.

CHECK THE ENDPOINT

Inductive impedance is +jωL; capacitive impedance is −j/(ωC).

T8 Complex impedance and series RL or RC

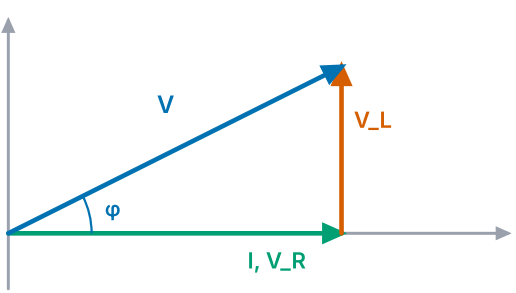
Purpose: The j operator keeps magnitude and phase together during circuit addition.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

The j operator keeps magnitude and phase together during circuit addition.



Series RL phasors with current as reference. V is the vector sum of V_R and V_L, so the source voltage leads current.

1. Use the common current as phasor reference in a series circuit.
   Why: The same current passes through each element.

2. Resistive voltage lies along I; inductive voltage is +90° and capacitive voltage −90°.
   Why: Use the pure-element phase laws.

3. Add perpendicular components, then use Pythagoras for magnitude.
   Why: Voltages with different phases do not add as scalar magnitudes.

4. The impedance angle is the voltage lead relative to current.
   Why: A positive angle means current lags the source.

RESULT

$Z_{RL} = R + j\omega L$; $Z_{RC} = R - j/(\omega C)$; $\tilde{I} = \widetilde{V}/Z$.

EXAM TRAP

Use j for the imaginary unit so i can denote current; phasors describe sinusoidal steady state at one frequency.

CHECK YOURSELF

A series RC circuit has negative impedance angle. Does current lead or lag?
Answer: It leads source voltage.

This topic supports Q15, Q16, Q32, Q33.

Work it through · Q15

Skill: Meaning of j. Model · syllabus-based

What does multiplying a phasor by j do, and why is j used rather than i here?

Since $j^2 = -1$, multiplication by j rotates a complex-plane vector +90° without changing its magnitude. The letter j avoids conflict with the time-dependent current i. It represents phase mathematically rather than a second physical current.

CHECK THE ENDPOINT

j denotes a +90° phase rotation.

Work it through · Q16

Skill: Phasor voltage addition. Model · syllabus-based

Why can magnitudes of resistor and inductor voltages not be added directly in a series RL circuit?

They differ in phase by 90°. With current as reference, $\widetilde{V} = V_R + jV_L$, so $V = \sqrt{V_R^2 + V_L^2}$, not $V_R + V_L$. Their instantaneous values still obey Kirchhoff's law.

CHECK THE ENDPOINT

Add voltages as phasors, then take the magnitude.

Work it through · Q32

Skill: Series RL phasor calculation. Model · syllabus-based

At an angular frequency where $X_L = 40$ Ω, a series RL circuit has R=30 Ω and source 100 V rms. Find Z, current, phase and real power.

1. $Z = 30 + j40$ Ω, so $|Z| = 50$ Ω.
   Why: Series impedances add as complex numbers.
2. $I_{\text{rms}} = 100/50 = 2$ A.
   Why: Use the magnitude form of Ohm's law.
3. $\phi = \tan^{-1}(40/30) = 53.13°$; current lags.
   Why: The impedance angle is positive.
4. $P = I^2R = 120$ W, also $VI\cos\phi = 100(2)(0.6)$.
   Why: Only R dissipates mean energy.

CHECK THE ENDPOINT

Z=30+j40 Ω; I=2 A lagging 53.13°; P=120 W.

Work it through · Q33

Skill: Series RC phasor calculation. Model · syllabus-based

A series RC circuit has R=60 Ω, $X_C = 80$ Ω and source 100 V rms. Find current, phase, component voltages and power.

1. $Z = 60 - j80$ Ω and $|Z| = 100$ Ω.
   Why: Capacitive impedance is negative imaginary.
2. I=1 A rms and $\phi = -53.13°$, so current leads source voltage by 53.13°.
   Why: The source voltage phase is measured relative to current.

3. $V_R = IR = 60$ V and $V_C = IX_C = 80$ V.
   Why: They are perpendicular phasors, whose resultant is 100 V.

4. $P = I^2R = 60$ W.
   Why: The ideal capacitor has zero mean power.

CHECK THE ENDPOINT

I=1 A leading 53.13°; $V_R = 60$ V, $V_C = 80$ V; P=60 W.

T9 Parallel LC and antiresonance

Purpose: Parallel branches share voltage; their reactive currents can cancel at the input.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

Parallel branches share voltage; their reactive currents can cancel at the input.

1. Add branch currents $I_C = j\omega CV$ and $I_L = V/(j\omega L)$.
   Why: Parallel elements see the same phasor voltage.

2. Factor V to identify admittance.
   Why: Current divided by voltage is Y.

3. At resonance the imaginary contributions cancel.
   Why: The branch currents have equal magnitudes and opposite phases.

4. The source current is zero although branch currents are not.
   Why: Current circulates between L and C.

RESULT

$Y = j(\omega C - 1/(\omega L))$; at $\omega_0 = 1/\sqrt{LC}$, ideal input Y=0 and input impedance diverges.

EXAM TRAP

This ideal parallel LC result differs from series resonance. Real coil resistance gives finite impedance and losses.

CHECK YOURSELF

At ideal parallel resonance, can inductor current be nonzero while source current is zero?

Answer: Yes; capacitor current cancels it at the input.

This topic supports Q17, Q34.

Work it through · Q17

Skill: Parallel resonance input. Model · syllabus-based

What are ideal input current and impedance at parallel LC resonance under an AC voltage source?

At $\omega_0 = 1/\sqrt{LC}$, $Y = j(\omega C - 1/\omega L) = 0$. Thus source current is zero and input impedance tends to infinity. Opposite nonzero branch currents can circulate.

CHECK THE ENDPOINT

Ideal parallel resonance gives zero input current and infinite impedance.

Work it through · Q34

Skill: Parallel LC branch currents. Model · syllabus-based

An ideal parallel LC has L=0.10 H and C=10 µF, driven at resonance by 10 V rms. Find frequency, branch currents and source current.

1. $\omega_0 = 1/\sqrt{LC} = 1000$ rad/s and $f_0 = 159.15$ Hz.
   Why: Set the input susceptance to zero.
2. At resonance $X_L = \omega_0 L = 100\ \Omega$ and $X_C = 1/(\omega_0 C) = 100\ \Omega$.
   Why: The two reactance magnitudes match.
3. $\tilde{I}_L = -j0.10$ A and $\tilde{I}_C = +j0.10$ A for voltage as reference.
   Why: Inductive current lags and capacitive current leads.

4. Their sum is zero.
   Why: Kirchhoff adds branch current phasors.

CHECK THE ENDPOINT

Branch currents are 0.10 A rms in opposite phase; ideal source current is zero.

T10 Series LCR and resonance

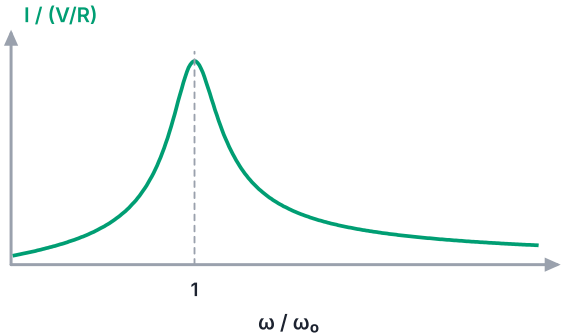
Purpose: In series, L and C reactances cancel at resonance, leaving only R.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

In series, L and C reactances cancel at resonance, leaving only R.



Computed series-resonance current curve for Q=5. Maximum current occurs at $\omega=\omega_0$; finite resistance keeps the peak finite.

1. Add the three impedances.
   Why: The same current flows through every element.
2. Set the imaginary part to zero for resonance.
   Why: Voltage and current become in phase.
3. Then impedance magnitude is minimum, R, and current is maximum for fixed source voltage.
   Why: The reactive contributions cancel as phasors.
4. Individual L and C voltages can be large even though their sum vanishes.
   Why: Equal opposite phasors need not be individually small.

RESULT

$Z = R + j(\omega L - 1/(\omega C))$; $\omega_0 = 1/\sqrt{LC}$; at resonance $I_{\mathrm{rms}} = V_{\mathrm{rms}}/R$.

EXAM TRAP

Below resonance the circuit is capacitive; above it the circuit is inductive.

CHECK YOURSELF

Does zero net reactance mean zero L and C voltages?

Answer: No; their nonzero phasors cancel.

This topic supports Q18, Q19, Q35, Q39.

Work it through · Q18

Skill: Series resonance condition. Model · syllabus-based

State the resonance condition and input impedance of a series LCR circuit.

Resonance occurs when $\omega L = 1/(\omega C)$, giving $\omega_0 = 1/\sqrt{LC}$. Reactive impedances cancel, so Z=R, current is in phase with voltage and its amplitude is maximal at fixed V.

CHECK THE ENDPOINT

At series resonance, Z=R.

Work it through · Q19

Skill: Above and below resonance. Model · syllabus-based

State whether a series LCR circuit is capacitive or inductive below and above its resonance frequency.

Below resonance, $1/(\omega C) > \omega L$, so net reactance is negative and current leads. Above resonance, $\omega L > 1/(\omega C)$, so net reactance is positive and

current lags.

CHECK THE ENDPOINT

Below: capacitive. Above: inductive.

Work it through · Q35

Skill: Series LCR resonance and voltage magnification. Model · syllabus-based

A series circuit has R=10 Ω, L=0.10 H and C=10 μF, with 10 V rms at resonance. Find frequency, current and L/C rms voltages.

1. $\omega_0 = 1000$ rad/s and $f_0 = 159.15$ Hz.
   Why: Use the LC resonance formula.
2. At resonance Z=R=10 Ω, so I=1 A rms.
   Why: Reactive drops cancel at the input.
3. Each reactance magnitude is 100 Ω, so $V_L = V_C = 100$ V rms.
   Why: Multiply branch reactance by the common series current.
4. These two voltages are opposite phasors and sum to zero, leaving $V_R = 10$ V.
   Why: A large internal voltage does not contradict the source voltage.

CHECK THE ENDPOINT

I=1 A; each reactive element carries 100 V rms although the source is 10 V rms.

Work it through · Q39

Skill: Series and parallel resonance comparison. Model · syllabus-based

Use complex impedance/admittance to explain and contrast series LCR resonance and ideal parallel LC resonance, including current and energy flow.

1. Series impedance is $Z_s = R + j(\omega L - 1/\omega C)$.
   Why: Common current permits impedance addition.

2. At $\omega_0 = 1/\sqrt{LC}$, $Z_s = R$, source current is maximum and in phase, and real power is V_{rms}^2/R.
   Why: Reactive voltage phasors cancel.

3. Individual L and C voltages may greatly exceed source voltage while cancelling each other.
   Why: Resonance constrains their sum, not individual values.

4. Parallel admittance is $Y_p = j(\omega C - 1/\omega L)$.
   Why: Common voltage permits admittance addition.

5. At the same ideal ω_0, $Y_p = 0$ and source current vanishes, while equal opposite branch currents remain.
   Why: Reactive current phasors cancel.

6. In both cases energy cycles between electric and magnetic stores; a real resistance causes loss and changes ideal sharpness.
   Why: Only dissipative elements consume mean power.

7. Specify the circuit and its losses before calling resonance “minimum” or “maximum” impedance.
   Why: Series and parallel results are opposite.

CHECK THE ENDPOINT

Series resonance minimizes impedance; ideal parallel resonance maximizes it, despite identical ideal LC frequency.

T11 Power in AC

Purpose: Only the part of current in phase with voltage transfers net energy each cycle.

BEFORE YOU START

Use P1 for the physical quantities and M1 for the mathematics. A capacitor obeys $q = Cv_C$ and an inductor obeys $v_L = L\,di/dt$ using passive signs. Current is positive into the positive-labelled capacitor plate, so $i = dq/dt$.

THE IDEA

Only the part of current in phase with voltage transfers net energy each cycle.

1. Write $v = V_0 \cos \omega t$ and $i = I_0 \cos(\omega t - \phi)$.
   Why: Define positive φ as current lag.

2. Multiply and apply the cosine product identity.
   Why: Separate a constant term from a double-frequency term.

3. Average over a whole cycle.
   Why: The oscillating term integrates to zero.

4. Replace peaks by rms values.
   Why: Each rms is peak divided by $\sqrt{2}$.

RESULT

$P = V_{\text{rms}} I_{\text{rms}} \cos\phi = I_{\text{rms}}^2 R$ for a series load with real resistance R.

EXAM TRAP

Apparent power VI in VA is not always real power in W. The power factor is cos φ.

CHECK YOURSELF

An ideal inductor draws rms current. Is its mean power necessarily positive?
Answer: No. Its phase angle is 90°, so mean power is zero.

This topic supports Q20, Q36, Q40.

Work it through · Q20

Skill: Power factor definition. Model · syllabus-based

Define power factor and distinguish real power from apparent power.

Power factor is $\cos\phi$, where φ is the phase difference of voltage and current for a sinusoidal load. Apparent power is $S_{\text{app}} = V_{\text{rms}} I_{\text{rms}}$ in VA; real mean power is $P = S_{\text{app}} \cos\phi$ in W.

CHECK THE ENDPOINT

Only the in-phase contribution transfers net energy per cycle.

Work it through · Q36

Skill: AC power numerical. Model · syllabus-based

A sinusoidal load takes 4 A rms from 230 V rms at power factor 0.75 lagging. Find real power, apparent power and phase angle.

1. Apparent power is $S_{\text{app}} = VI = 230(4) = 920$ VA.
   Why: This is the product of rms magnitudes.
2. Real power is $P = VI\cos\phi = 920(0.75) = 690$ W.
   Why: Only the in-phase part contributes on average.
3. $\phi = \cos^{-1}(0.75) = 41.41^\circ$ with current lagging.
   Why: The sign/lead description comes from the supplied power-factor qualifier.

CHECK THE ENDPOINT

P=690 W, apparent power=920 VA, lag angle≈41.41°.

Work it through · Q40

Skill: Derivation of average AC power. Model · syllabus-based

Derive average power for sinusoidal voltage and current with phase difference, then discuss pure R, L, C and series LCR at resonance.

1. Let $v = V_0 \cos \omega t$ and $i = I_0 \cos(\omega t - \phi)$.
   Why: φ is the voltage lead over current.

2. Use $2 \cos A \cos B = \cos(A - B) + \cos(A + B)$ to obtain $p = (V_0 I_0/2)[\cos \phi + \cos(2\omega t - \phi)]$.
   Why: Power is the instantaneous product vi.

3. Average over a full cycle: the second cosine contributes zero.
   Why: It completes an integer number of cycles.

4. Since $V_{\text{rms}} = V_0/\sqrt{2}$ and $I_{\text{rms}} = I_0/\sqrt{2}$, $P = V_{\text{rms}} I_{\text{rms}} \cos \phi$.
   Why: Convert peaks to rms values.

5. For pure R, φ=0 and P=I²R; for pure L or C, φ=±π/2 and P=0.
   Why: Reactive elements return the energy they store in the ideal model.

6. At series LCR resonance φ=0, so all input mean power is dissipated in R, although reactive voltages and stored energies are nonzero.
   Why: Cancellation of reactances gives unity power factor.

CHECK THE ENDPOINT

Average AC power equals rms voltage × rms current × power factor.

Formula and condition sheet

T1 · LR current growth

$i = (V/R)(1 - e^{-t/\tau_L})$, $\tau_L = L/R$, for i(0)=0 and a DC step V.

T2 · LR current decay

$i = I_0 e^{-tR/L}$; $U_L = LI_0^2 e^{-2tR/L}/2$.

T3 · RC charging and discharging

Charging: $q = CV(1 - e^{-t/RC})$, $i = (V/R)e^{-t/RC}$. Discharging: $q = q_0 e^{-t/RC}$, $i = -q_0 e^{-t/RC}/(RC)$ with i=dq/dt.

T4 · Free LC oscillations

$\ddot{q} + q/(LC) = 0$; $\omega_0 = 1/\sqrt{LC}$; for q(0)=q_0, i(0)=0: $q = q_0 \cos\omega_0 t$, $i = -\omega_0 q_0 \sin\omega_0 t$.

T5 · LC response to a DC step

For initially uncharged C and zero current: $q = CV(1 - \cos\omega_0 t)$, $i = CV\omega_0 \sin\omega_0 t$.

T6 · Rotating-coil AC generator

$\mathcal{E} = NBA\omega \sin\omega t$, peak $\mathcal{E}_0 = NBA\omega$, when the coil normal starts along B.

T7 · Pure R, L and C on AC

$Z_R = R$, $Z_L = j\omega L$, $Z_C = 1/(j\omega C) = -j/(\omega C)$.

T8 · Complex impedance and series RL or RC

$Z_{RL} = R + j\omega L$; $Z_{RC} = R - j/(\omega C)$; $\tilde{I} = \widetilde{V}/Z$.

T9 · Parallel LC and antiresonance

$Y = j(\omega C - 1/(\omega L))$; at $\omega_0 = 1/\sqrt{LC}$, ideal input Y=0 and input impedance diverges.

T10 · Series LCR and resonance

$Z = R + j(\omega L - 1/(\omega C))$; $\omega_0 = 1/\sqrt{LC}$; at resonance $I_{\text{rms}} = V_{\text{rms}}/R$.

T11 · Power in AC

$P = V_{\text{rms}} I_{\text{rms}} \cos\phi = I_{\text{rms}}^2 R$ for a series load with real resistance R.

Past-paper route

No question from the supplied local past papers maps directly to this module. All 40 are model questions, guided by the syllabus; no invented PYQ provenance.